Introduction to Wavelets

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Key Concepts

1) Wavelets are bases of varying frequency and duration.

    a) Wavelets can localize signals in both time and frequency.

    b) A large class of signals can be represented with relatively few bases.

    c) Fast algorithms exist for computing wavelet expansions.

2) Wavelet representations are most easily developed for continuous-time signals.

3) The wavelet basis for scale (frequency) j and time (location) index k is obtained by scaling (by 2^j) and time shifting (by k) a mother wavelet.

4) Scaling by 2^j in time causes the inverse scaling (by 2^{-j} in the frequency domain.

    a) Increasing j by one halves the duration and doubles the bandwidth.

    b) Very short duration wavelets (large j) have fine time resolution, but coarse frequency resolution. This allows fine localization in time of transient events.

c) Very long duration wavelets (small j) have coarse time resolution, but fine frequency resolution. This allows fine localization in frequency of long duration events.

5) The short-time Fourier transform uses fixed duration bases and thus has constant time and frequency resolution.

6) Continuous-time basis expansions are analogous to finite-dimensional (vector) basis expansions.

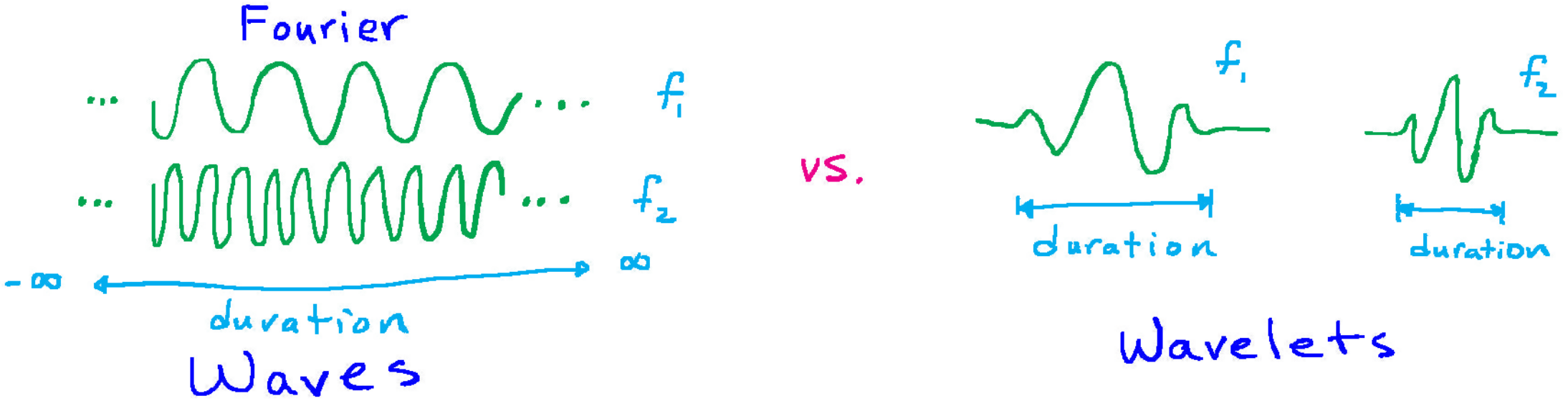
a) The inner product in continuous time is defined as the integral of the product of two signals.

b) Orthonormal bases have unit energy and the inner product between different bases is zero.

c) The wavelet basis coefficients are the inner product of the wavelet with the signal when the wavelets are orthonormal.

Video Screenshots

Wavelets - bases of varying "frequency" and duration



1. Can localize signals in **both** time and frequency
2. A large class of signals can be represented with relatively few bases
3. Efficient (fast) algorithms exist for computing coefficients

Wavelet bases are described by a pair of indices

k: time (location) index ; j: frequency index

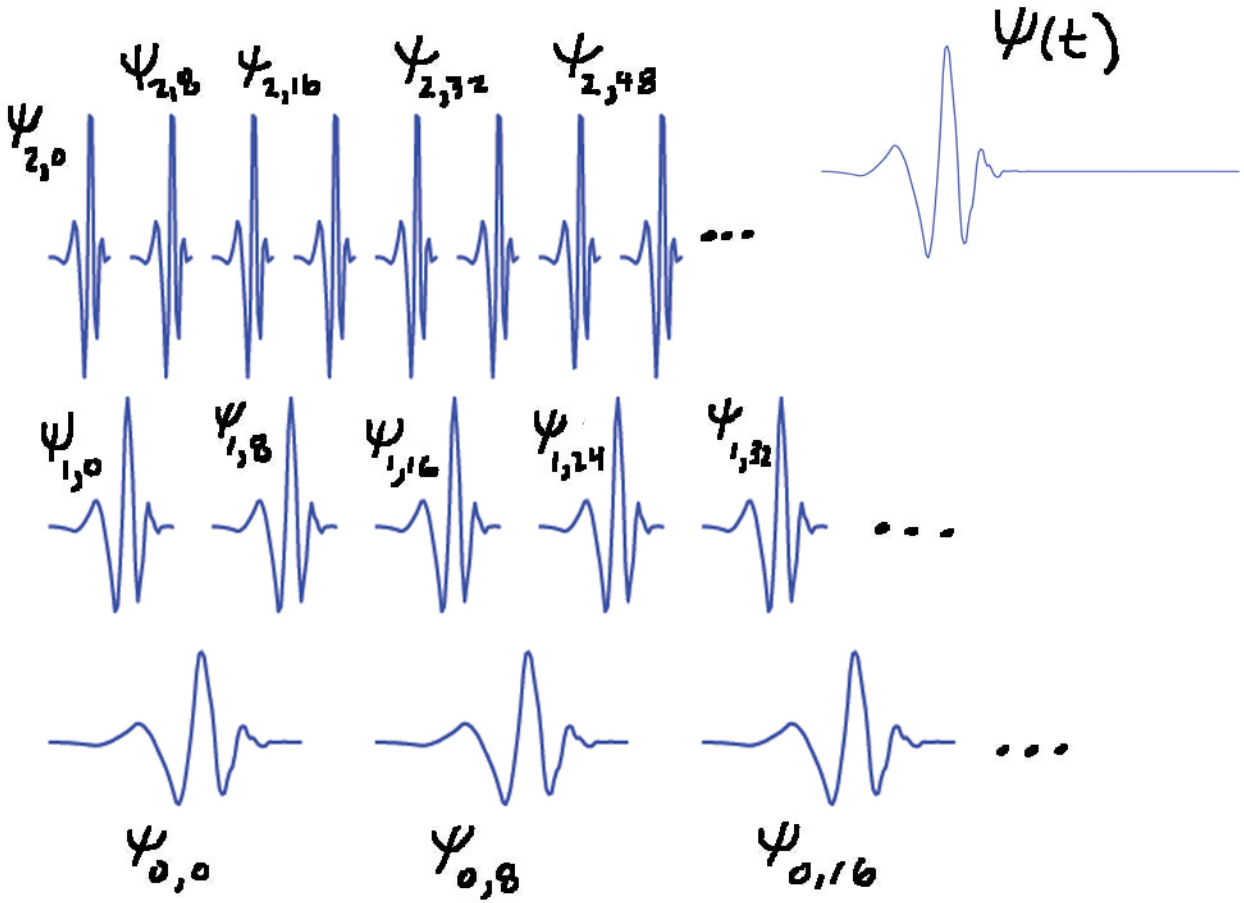
$$x(t) = \sum_j \sum_k a_{j,k} \underbrace{\psi_{j,k}(t)}_{\text{basis}}$$

- continuous-time is more natural

Mother wavelet: $\psi(t)$

$$\psi_{j,k}(t) = 2^{j/2} \psi(2^j t - k)$$

- $2^{j/2}$: energy norm.
- 2^j: time scaling
- $-k$: time shift

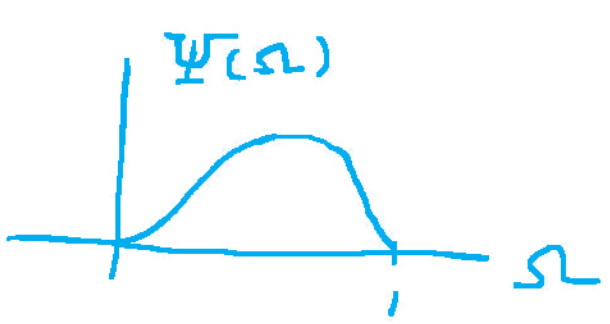


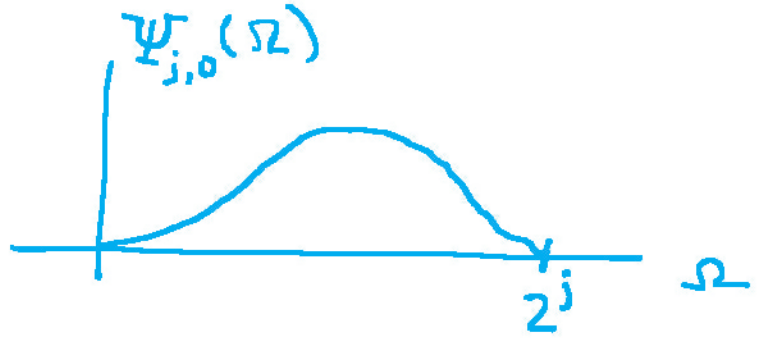
Scaling in the frequency domain

$$\psi(t) \xleftrightarrow{FT} \Psi(\Omega)$$

$$\psi_{j,0}(t) = 2^{j/2}\psi(2^j t) \xleftrightarrow{FT} \Psi_{j,0}(\Omega) = 2^{-j/2}\Psi(2^{-j}\Omega)$$

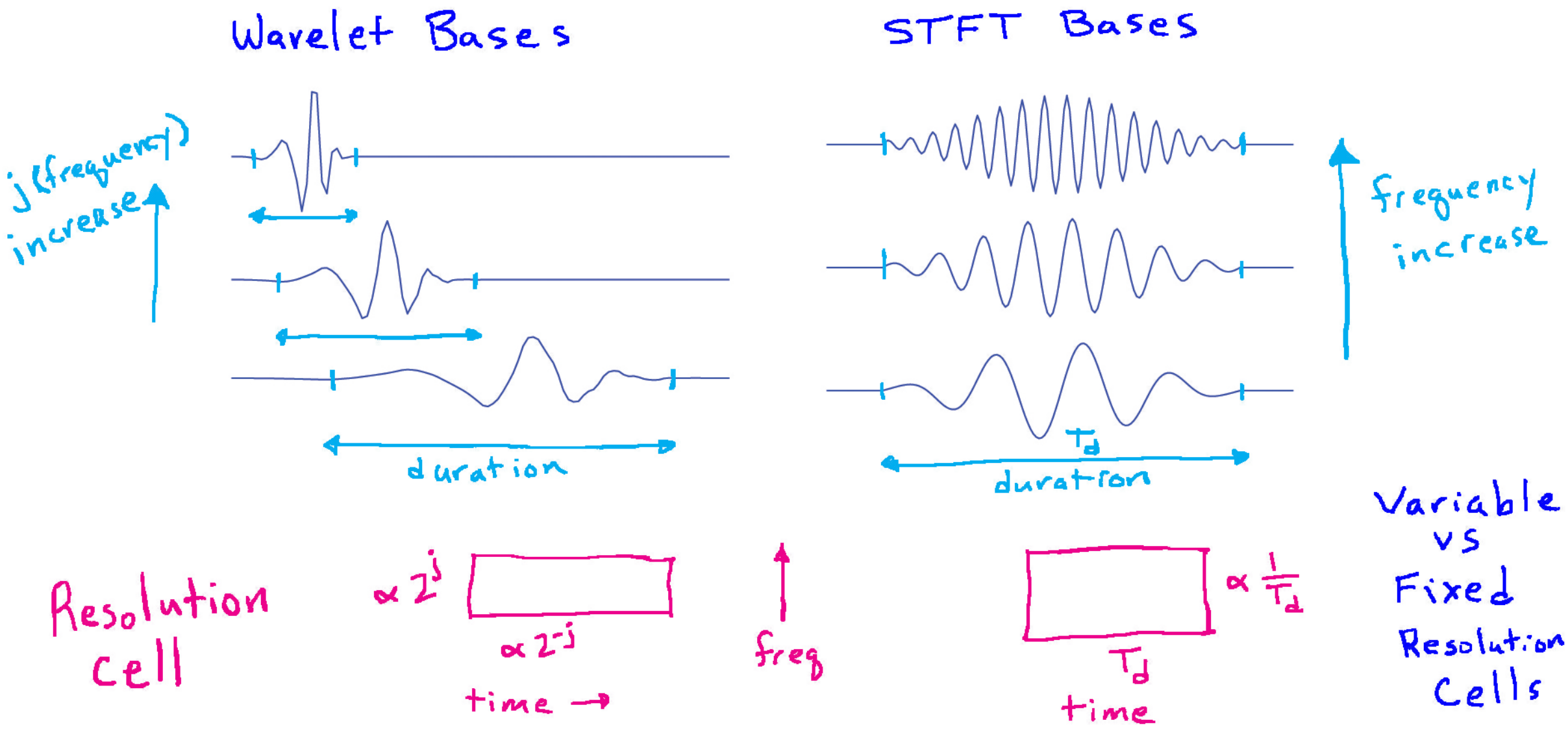
$j>1$: compress $\xleftrightarrow{FT}$ expand





Increasing j by 1 doubles the bandwidth and halves the duration

Wavelets vs. Short-time Fourier transform



Working with continuous time bases

Define inner product $\langle f(t), g(t) \rangle = \int f(t) g(t)\, dt$

Orthonormal wavelets

$$\langle \psi_{j,k}(t), \psi_{\ell,m}(t) \rangle = \int \psi_{j,k}(t)\, \psi_{\ell,m}(t)\, dt = \begin{cases} 0 & j \neq \ell \text{ or } k \neq m \\ 1 & j = \ell \text{ and } k = m \end{cases}$$

Wavelet expansion

$$x(t) = \sum_j \sum_k a_{j,k}\, \psi_{j,k}(t)$$

if o/n basis

$$a_{j,k} = \langle \psi_{j,k}(t), x(t) \rangle = \int \psi_{j,k}(t)\, x(t)\, dt$$

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